Improving Spatial Generalization with Edge-Gated Spatiotemporal Graph Neural Networks for Temperature Inversion Forecasting

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Abstract

Near surface temperature inversion detection and forecasting is key for tree fruit crop production management, especially during spring frost season. Inversions also play a critical role in decisions related to efficient crop protection with minimal pesticide drift. For farmers to proactively alleviate the consequences of spring frost damage, accurate forecasts of local temperature inversion strengths for next 24 to 72 h is crucial. As typical weather stations quantify surface aggregate conditions at 1.5 m, estimating 9 m air temperature and inversion strength (i.e., difference between 9 m and 1.5 m air temperature) at such locations is a challenging problem in atmospheric science with agricultural applications. This work introduces a spatiotemporal Graph Neural Network model that learns to generalize to unseen locations by modeling interactions between nearby mesonet tower weather stations with 9 m air temperature sensors using terrain-derived edge features. We evaluate our model on tower station measurements from the AgWeatherNet in Washington State, and compare against several baseline algorithms. We test spatial generalization through two settings of spatial cross-validation: leave-one-station-out and clustering. Our model demonstrates improved performance in both settings over the baselines, and a small set of ablations quantifies the contributions of specific components of our model. Our model is being validated against unseen tower station data to test its robustness and ensuing application to non-tower stations, and will be incrementally made available to growers through AgWeatherNet's decision support portal. Inversion forecasts for the next 48 h, particularly on non-tower stations, will aid in effective frost risk mitigation and drive crop protection spray applications, improving resource use efficiency in fresh market tree fruit and berry crop production management.

1 Introduction

Temperature inversion is a meteorological phenomenon that describes a layer in the atmosphere in which temperature increases with altitude. Our work focuses on forecasting temperature inversions near the surface level - up to 9 meters above ground - in contrast to inversions that take place higher up in the atmosphere. These near surface inversions are measurable in near real-time locally using mesonet tower type weather stations, such as those deployed by AgWeatherNet (AWN) across Washington State (Amogi, Khot, and Hill 2024; Amogi, Khot, and Hoheisel 2025). Due to deployment and maintenance costs, regional networks including AgWeatherNet often deploy a blend of mesonet and typical surface aggregate stations that only quantify weather at 1.5 m above ground. For example, of over 370 stations only 61 are mesonet tower stations in the current AWN network. Besides regional networks, fruit crop growers often deploy similar low-cost surface aggregate stations that fail to quantify inversions as they lack the 9 m air temperature sensor.

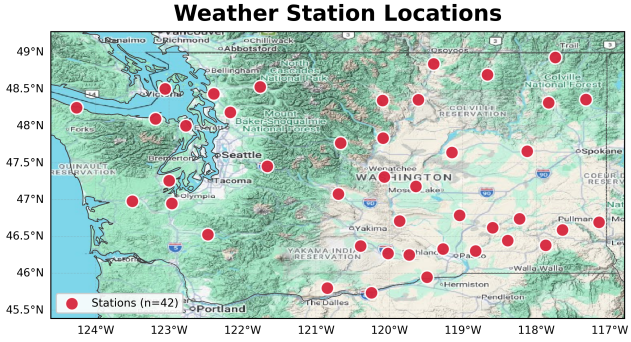


Figure 1: Mesonet tower weather station locations across Washington State, each station shown as a red dot.

Accurately forecasting air temperature inversions in local areas is essential in agriculture. Adverse weather, including inversions, has a range of negative effects on farm production, such as frost formation on crops when surface air temperature drops below zero (Bates 1971; Bates, Lombard et al. 1978; Ohlemeier 2025; Snyder and Melo-Abreu 2005) and drift of hazardous crop protectant pesticide sprayers to non-target areas (Bish, Guinan, and Bradley 2019), and contributing to air quality degradation (Rendón et al. 2015; Wallace, Corr, and Kanaroglou 2010; Wang et al. 2009; Xu et al. 2019) and poor visibility (Dennstaedt 2023). Access to localized inversion forecasts allows farmers to do resource-use efficient crop protection. For example, timely actuation of wind machines when there is a strong inversion can prevent frost damage in fruit and berry crops by pooling warmer air from the top layer (Bates 1971; Bates, Lombard et al. 1978; Amogi, Khot, and

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Hill 2024; Khot et al. 2026; Prengaman 2025), and avoiding pesticide spray until the inversion is suitably weak (Amogi et al. 2025). AgWeatherNet provides frost risk assessments and pesticide spray decision support to farmers.

Temperature inversions exhibit strong seasonal behavior, being more common and generally stronger in the overnight hours and colder months (Abdul-Wahab et al. 2004; Kahl, Serreze, and Schnell 1992; Noad et al. 2023; Riordan, Davis, and Kiess 1986; Rupp et al. 2020). Inversion prevalence varies by local topography (Daly, Conklin, and Unsworth 2009; Rupp et al. 2020), and cold air often flows from higher to lower elevations pooling in basins (Rupp et al. 2020). Regionally, these pools create patterns that influence the directional spread of inversions into and within basins.

It is desirable to provide inversion measurements and forecasts across the major crop production areas. Prior efforts on inversion forecasting largely focus on the tower stations and there is lack of literature on forecasting inversions for non-tower stations. Our work aims to improve the spatial generalization capacity of inversion predictions, while also allowing for predictions at new stations, unseen regions, and step towards forecasts on non-tower stations. We propose a spatiotemporal Graph Neural Network (GNN) customized to this task that utilizes terrain-based node and edge features within immediate spatial neighborhoods along with recurrent neural networks for forecasting temperature inversions locally. We emphasize forecast persistence through a "persistence anchoring" formulation in the temporal portion of our model inspired by the meteorological observations of inversion persistence (Largeron and Staquet 2016; Rupp et al. 2020). For brevity, we address this work as EGP-GNN for **E**dge-**G**ated, **P**ersistence-aware **G**raph **N**eural **N**etwork.

2 Related Work

Existing work on temperature inversions falls broadly into two categories: analysis and prediction. There exists a relatively much larger body of work on the analysis of inversions; which involves detection, characterizing inversion properties, and mitigating the impact of inversions. Inversion measurement and detection is often done at a higher vertical scale than our work, using radiosonde measurements to produce vertical profiling of temperature using weather balloons (Abdul-Wahab et al. 2004; Joly and Richard 2022a,b; Lagmiri and Dahech 2024; Rendón et al. 2014; Rendón et al. 2015; Xu et al. 2021; Zhang, Zhang, and Yang 2022; Zhang et al. 2025b,a). Other work uses aerial drone based weathersondes (Dadallage et al. 2026) or remote sensing data for detecting inversions and related weather conditions (Adolph, Albert, and Hall 2018; Collados Lara et al. 2020; Kachar et al. 2015; Liu and Key 2003). In our case, local weather station networks are used to detect or categorize inversion presence and strength at individual stations (Amogi, Khot, and Hoheisel 2025; Battany, Smith, and Snyder 2025; McCaffrey et al. 2019; NDSU 2021; Rupp et al. 2020). Closely related is work on cold air pooling analysis (Dorninger et al. 2011; Lareau et al. 2013; McCaffrey et al. 2019; Sheridan 2019; Whiteman et al. 2000; Zängl 2005), where cooled air pools in valleys and basins and produces inversions.

Past forecasting work primarily deals with elevated inversions or atmospheric characteristics, such as estimating the planetary boundary layer height (PBLH) (Christopoulos et al. 2025; Haby 2024; Seibert et al. 2000; Silva et al. 2025), which can be used as an indirect indicator of inversions on a larger regional scale. Of note is the work from (Su and Zhang 2024) which predict the PBLH from conventional meteorological measurements using deep learning. A survey by (Canché-Cab et al. 2024) categorizes various applications of machine learning on the PBLH.

Local, near surface temperature inversion forecasting predicts low-level inversions at localized points, which is important for agriculture. To our knowledge, few works have addressed forecasting at this scale on individual weather stations. The AgWeatherNet system provides real-time forecasts for inversion strengths using mesonet tower weather station measurements (Amogi, Khot, and Hoheisel 2025; Khot et al. 2026). The most closely related deep learning approach uses terrain features and past history in a Long Short-Term Memory model to improve inversion forecasting within Yakima Valley in Washington State (Dinkins et al. 2026). These approaches focus on learning temporal patterns to forecast inversion strength, in contrast to spatial generalization, which is focus of this study. Since inversion presence and movement are influenced heavily by topography, learning spatial relationships across nearby weather stations has the potential to improve local forecasting and spatial generalization.

Graph Neural Networks are a type of neural network designed for data represented as a graph of nodes and edges (Wu et al. 2020a). GNNs have been applied to social networks (Hamilton, Ying, and Leskovec 2017), recommender systems (Ying et al. 2018), molecular and protein graphs (Gilmer et al. 2017), and time series forecasting (Jin et al. 2024; Wu et al. 2020b). Spatiotemporal GNNs exist in areas such as traffic forecasting (Bai et al. 2020; Li et al. 2017; Yu, Yin, and Zhu 2017; Wu et al. 2019), regional weather forecasting (Lam et al. 2023), and kriging (Islam, Tadepalli, and Fern 2026; Wu et al. 2021). Representing weather station networks is a natural application of GNNs, with stations as nodes, and edge connections as a spatial or topographic relationship between locations.

Many GNN architectures have been proposed in the literature. We include some relevant examples for our work. Graph Convolutional Networks (GCNs) perform aggregation over local neighborhoods (Gilmer et al. 2017; Kipf and Welling 2016) through a form of "message passing". These extend to edge-conditioned message passing using features on the edges between nodes to inform the aggregation (Gilmer et al. 2017; Simonovsky and Komodakis 2017). Graph Attention Networks extend GCNs to learn attention weights that determine the relative importance of nodes in local neighborhoods (Brody, Alon, and Yahav 2021; Veličković et al. 2017). Learnable gating mechanisms exist to adaptively modulate the flow of information in the GNNs (Bresson and Laurent 2017; Jiao et al. 2021; Li et al. 2015). Spatiotemporal GNNs pair spatial aggregation functions with temporal forecasting models, such as temporal convolutional networks (Lea et al. 2017; Wu et al. 2019) or recurrent neural networks (Bai et al. 2020; Jin et al. 2024; Li et al. 2017).

Table 1: Dynamic inputs. Recorded weather variables are observed over a 24 h history window; future-known forecast covariates are applied over the full 48 h forecast horizon. We use three seasonal variables for time-of-day and yearly effects, along with the solar zenith angle, which is a deterministic function of the station location and timestamp.

Variable	Units	History	Forecast
Air temperature (1.5 m)	°F	•	
Dewpoint temperature	°F	•	
Relative humidity	%	•	
Solar radiation	W/m^2	•	
Mean wind speed	mph	•	
Maximum wind gust	mph	•	
Precipitation	in	•	
Soil temperature (2 in and 8 in)	°F	•	
Hour of day	h	•	•
Day of year	d	•	•
Month of year	mo	•	•
Solar zenith angle	–		•

3 Methodology

3.1 Data

Our primary data consists of historical weather variable measurements from the AgWeatherNet (AWN) weather station network in Washington State. We focus on 42 tower stations (see Figure 1) spread across the state with consistent readings at 15-min intervals without large gaps in coverage or high percentages of erroneous measurements.

First, let $\mathbf{x}_i(t)$ be the recorded weather sensor variables for station i at time t. These dynamic weather variables, along with several time-specific features, are presented in Table 1. Near surface temperature inversion strengths $\delta_i(t, \theta)$ at these stations and scale are defined as the difference between the 9 meter air temperature sensor reading $\theta_i^{9\mathrm{m}}(t)$, and the 1.5 meter air temperature sensor reading $\theta_i^{1.5\mathrm{m}}(t)$. Formally,

$$\delta_i(t, \theta) = \theta_i^{9\mathrm{m}}(t) - \theta_i^{1.5\mathrm{m}}(t) \tag{1}$$

We model the measured inversion strength $\delta_i(t, \theta)$ as the response variable. Our aim is to forecast at any stations present in Washington State, even non-tower stations. This motivates the need for spatial generalization during modeling. We cannot evaluate the models in the absence of ground truth $\theta_i^{9\mathrm{m}}(t)$ measurements, thus we are limited to the use of tower stations here for model development and validation.

We augment with a set of static features covering geographic position and terrain for each station. These are divided into two sections: station-level features at any given station i, and edge features between neighboring stations $j \in \mathcal{N}(i)$, where $\mathcal{N}(i)$ denotes the neighbors of station i. Station-level features cover the location and profiling for a station and its immediate region. Edge features capture the relationship between a weather station and its closest neighbors by coordinates. A comprehensive summary of these features is presented in Table 2. All terrain features are computed from Digital Elevation Models (DEMs) collected from the US Geological Survey (USGS) at 10 meter resolution (United States Geological Survey 2021). Land cover features are calculated as a histogram from the National Land Cover Database 2023 (NLCD) (Survey 2024).

The tower stations from AWN have a variable history, based on the installation dates of the stations. We use all available data from these 42 stations, which have at least 1 year of measurements, starting from July 2021, up through August 2025, though many stations do not have coverage for the entire period. The weather readings we used were 1 h summaries generated from 15-min interval measurements.

We pre-process this data by linearly imputing any missing values and perform quality control in line with existing work (Dinkins et al. 2026) to correct obvious out-of-bounds measurements and sentinel values, and z-normalize the weather variables and edge features using training data.

3.2 Model Definition

We introduce a spatiotemporal GNN model to use inherent spatial relationships between the stations in Washington State, which we label EGP-GNN .

Let $t \in [T - 24, T)$ cover the 24 hour history window of sequential measurements of the dynamic features, and $\tau \in [T, T + 48)$ cover the 48 hour forecast horizon, for any time T in the data. We include $\cos Z_i(\tau)$ as listed in Table 1, which is the cosine of the solar zenith angle (Spencer 1971), calculated as a function of the time of day, day of year, and the latitude. We construct a future set of known time-dependent covariates $\mathbf{u}(\tau)$ by concatenating the forecast features from Table 1, covering month/day/hour and $\cos Z_i(\tau)$. These help capture the daily and yearly cyclical patterns of temperature inversions. We augment the station-level static features $\mathbf{s}_i$ from Table 2 with a representation from a convolutional neural network (CNN) autoencoder (Hinton and Salakhutdinov 2006; LeCun et al. 1989; LeCun, Bengio, and Hinton 2015) trained on the local (4 km x 4 km) DEM files $\mathbf{D}_i$ centered at each station, respecting data splits over each fold, containing elevation at 10 m resolution. The CNN captures a latent representation of the terrain, rather hand-engineered features, that we posit may prove useful for forecasting inversions.

$$\tilde{\mathbf{s}}_i = [\mathbf{s}_i;\, \mathrm{CNN}(\mathbf{D}_i)] \tag{2}$$

We perform a series of spatial aggregation steps to encode, at each history window time step t, the graph state for a node i given the neighbors $j \in \mathcal{N}(i)$. We let $|\mathcal{N}(i)| = 3$ for this work. Note that empirically adding higher numbers of neighbors ($|\mathcal{N}(i)| > 3$) degrades performance, as the spatial distances become larger and relationships between the stations no longer informative for temperature inversions.

First, the edge features $\mathbf{e}_{ij}$ we pass through a two layer multi layer perceptron neural network (MLP) to get a learned edge embedding $\mathbf{p}_{ij}$ that represents a learned spatial relationship between stations i and j. We apply a "message" two layer MLP to the weather variables, the neighbors' augmented static features, and the edge embedding.

$$\mathbf{p}_{ij} = \mathrm{MLP}_e(\mathbf{e}_{ij}), \tag{3}$$

$$\mathbf{m}_{ij}(t) = \mathrm{MLP}_m\big([\mathbf{x}_j(t);\, \tilde{\mathbf{s}}_j;\, \mathbf{p}_{ij}]\big). \tag{4}$$

Table 2: Static inputs, both station-level and edge features between nearest-neighbors. Station-level features are summary statistics either at the station's point or aggregated over a neighborhood (4 km x 4 km), as in existing work (Dinkins et al. 2026). Edge features describe the relationship between a target station and each of its neighbors, and are computed by sampling at 1km intervals along a straight line between any two stations' latitude/longitude values.

Family	**#**	**Description**	**Units**
Station-level features $\mathbf{s}_i \in \mathbb{R}^{37}$			
Position	3	Latitude, longitude, and elevation of the station	deg, deg, ft
Terrain (center pixel)	7	Slope, aspect, curvature, and hillshade at the tower pixel	deg, %, index
Terrain (neighborhood median)	7	Same seven descriptors, median-aggregated over the surrounding patch	deg, %, index
Land cover	20	NLCD class fractions within the station neighborhood	fraction
Edge features $\mathbf{e}_{ij} \in \mathbb{R}^{40}$			
Path geometry	3	Bearing (clockwise from DEM grid north), sampled length, and profile sample count of the $i \to j$ edge	deg, km, count
Elevation profile	6	Max/min, range, std deviation, skewness, and relief ratio of the elevation profile along the path	m, -
Barrier	3	Height of the maximum intervening ridge from each endpoint, and the difference between them	m
Path slope and roughness	5	Mean/max/std dev of the slope along the path, concavity fraction, and mean local terrain ruggedness along the path	deg, fraction
Endpoint features	16	Elevation, ruggedness, and topographic position and relative elevation % (at three scales: 300 m, 1 km, and 4 km) of each endpoint	m, fraction
Endpoint feature differences	6	Signed $i - j$ differences in topographic position relative elevation % at each of the three scales	m, fraction

We calculate attention weights with another two layer MLP for each neighbor j, as well as a gating mechanism with a single layer MLP, that collectively determine the weight on each neighbor's message $\mathbf{m}_{ij}(t)$, and which portions of this message are relevant for predictions. These are defined as

$$\alpha_{ij}(t) = \underset{j}{\text{softmax}}\ \text{MLP}_\alpha\big([\mathbf{x}_i(t);\, \mathbf{x}_j(t);\, \tilde{\mathbf{s}}_i;\, \tilde{\mathbf{s}}_j;\, \mathbf{p}_{ij}]\big), \tag{5}$$

$$\mathbf{g}_{ij}(t) = \text{MLP}_g\big([\mathbf{x}_j(t);\, \mathbf{p}_{ij}]\big) \tag{6}$$

We aggregate the messages into a spatial embedding $\hat{\mathbf{z}}_i(t)$ with message passing, and concatenate the aggregates with a weighted self-channel, both of which have LayerNorm (LN) applied (Ba, Kiros, and Hinton 2016). This yields the combined spatial embedding for time t, $\boldsymbol{\zeta}_i(t)$ as

$$\hat{\mathbf{z}}_i(t) = \text{LN}\Big(\textstyle\sum_{j\in\mathcal{N}(i)} \alpha_{ij}(t)\, \mathbf{g}_{ij}(t) \odot \mathbf{m}_{ij}(t)\Big) \tag{7}$$

$$\boldsymbol{\zeta}_i(t) = \big[\hat{\mathbf{z}}_i(t);\, \text{LN}\big(W_{\text{self}}\, \mathbf{x}_i(t)\big)\big]. \tag{8}$$

From here, we apply a sequence-to-sequence style encoder-decoder model (Sutskever, Vinyals, and Le 2014) for the temporal forecasting. We calculate a historical embedding of the graph $\mathbf{H}_i^{\text{enc}}$ by passing each spatial embedding $\boldsymbol{\zeta}_i(t)$ through a recurrent neural network encoder, in this case a Gated Recurrent Unit (GRU) (Chung et al. 2014).

$$\mathbf{H}_i^{\text{enc}} = \text{GRU}_{\text{enc}}\big(\boldsymbol{\zeta}_i(0), \ldots, \boldsymbol{\zeta}_i(T-1)\big) \tag{9}$$

For each forecast horizon τ, we autoregressively apply a GRU decoder network to calculate an output and hidden state tuple to the future time covariates $\mathbf{u}(\tau)$ and the previous time's target (or output) value $\tilde{y}_i(\tau - 1)$

$$\big(\mathbf{o}_i(\tau),\, \mathbf{h}_i(\tau)\big) = \text{GRU}_{\text{dec}}\big([\tilde{y}_i(\tau-1);\, \mathbf{u}(\tau)],\, \mathbf{h}_i(\tau-1)\big). \tag{10}$$

Since temperature inversions can persist for extended periods (Rupp et al. 2020), we add a learnable "persistence anchoring" parameter $\lambda_i(\tau)$ to prevent forecasts from drifting substantially over time. This balances between the value at the previous time step and the forecast value at the current time. The decoder operates autoregressively with linearly decaying teacher forcing during training. Similar logic is applied to the anchoring and the value $\tilde{y}_i(\tau)$, which starts from the last observed value of the history $y_i(T-1)$. In the absence of this observed value, such as for non-tower stations, we can fall back to predicting or seeding this value. During rollouts after the initial value, the model anchors on the previous autoregressive prediction when teacher forcing is absent.

$$\lambda_i(\tau) = \sigma\big(\text{MLP}_\lambda\big([\mathbf{o}_i(\tau);\, \mathbf{u}(\tau)]\big)\big), \tag{11}$$

$$a_i(\tau) = \begin{cases} y_i(T-1), & \tau = 1, \\ \tilde{y}_i(\tau-1), & \tau > 1, \end{cases} \tag{12}$$

$$\hat{y}_i(\tau) = \lambda_i(\tau)\, a_i(\tau) + \big(1 - \lambda_i(\tau)\big)\big(\mathbf{w}_o^\top \mathbf{o}_i(\tau) + b_o\big). \tag{13}$$

We present a comparison against several baseline algorithms. First is the Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber 1997) algorithm from (Dinkins et al. 2026). Next, we add a Gaussian Process head to the

LSTM model using GPyTorch (Gardner et al. 2018; Williams and Rasmussen 2006), following the deep kernel learning (DKL) literature (Ober, Rasmussen, and van der Wilk 2021; Wilson et al. 2016), labeled DKL-LSTM. For DKL-LSTM, we pretrain the LSTM before training the GP parameters, then fine-tune both in tandem after a set number of epochs. Finally, we compare against GraphWaveNet (Wu et al. 2019), a widely used spatiotemporal GNN that is a strong baseline. We give all models the same station-level static feature set, the CNN derived embeddings, and the forecast variables from Table 1. The weather stations vary by installation date, and thus the graph itself is time-varying, both in nodes and edge connections. We train with root mean-squared error (RMSE) as the loss function.

Training data comprises all possible time-ordered sequences from available measurements from 2021-07-15 to 2024-06-01, validation from 2024-06-04 to 2025-01-15, and test from 2025-01-19 to 2025-08-29. All models perform forecasts using a 24 h history window and generate hourly forecasts out to a horizon of 48 h. During training, because of a large number of sequences, many of which have overlapping timestamps, we downsample stochastically to 25% of the total available training sequences during each epoch using a fixed random seed shared across all models. Hyperparameters settings are found in the Appendix along with station information. Given the dynamic nature of the graph, we run GraphWaveNet inductively on per-sample ego graphs (target with 3-nearest-neighbors) with a fixed star adjacency within each sample and adaptive adjacency disabled. Adaptive adjacency is learned from persistent node embeddings off of a fixed graph, which cannot be applied in our dynamic graph context. The validation set RMSE is used for model selection and tuning via grid search on the cluster-based cross validation in Section 4.2.

The persistence anchoring value $\lambda_i(\tau)$ uses $y_i(T-1)$ to seed the anchoring. On tower stations, this is available at the target station and can be used, thus the EGP-GNN experiments represent a performance ceiling when the seeding is perfect. On non-tower stations, this value must be estimated, either from EGP-GNN as an extra prediction or from imputation. We include EGP-GNN (RF seed) where a seed comes from a simple pretrained Random Forest (RF) (Breiman 2001) imputation model to predict $y_i(T-1)$ that respects identical data splits and does not use a non-tower target station's inversion strength. This RF is used off-the-shelf and is viewed as a floor for the seeding performance.

4 Results and Discussion

Cross validation (CV) in a spatial setting requires consideration for how the model will be used, along with avoiding schemes that can bias performance estimates (e.g. spatial autocorrelation between neighboring locations) (Koldasbayeva and Zaytsev 2025; Mahoney et al. 2023; Roberts et al. 2017; Wang et al. 2023). Following best practices for matching our setting (Roberts et al. 2017; Wang et al. 2023), we consider both leave-one-station-out and cluster-based cross-validation for our analysis. Leave-one-station-out cross-validation evaluates how well models generalize to new individual stations within a larger region, whereas cluster-based CV holds out distinct groups of stations within different regions and measures regional transfer. For both settings, we respect data splits for station neighbor definitions, while allowing validation stations to access training stations, and testing stations to access both. This mimics unseen stations being added to an evolving graph. During training, stations define their neighbor set from training stations that were installed prior to the first reading of any historical sequence. This ensures consistent neighbors during a sample for forecasting.

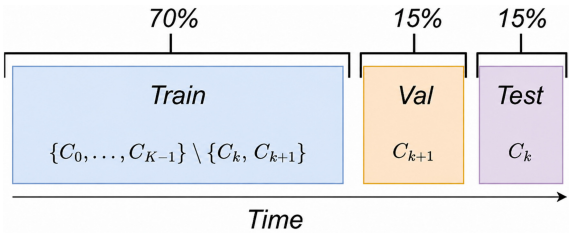


Figure 2: The cluster based cross-validation data split ($K = 6$). Clusters have no overlap. The validation cluster C_{k+1} is used for tuning and model selection.

4.1 Leave-One-Station-Out

To measure the models in the situation in which a single new station is installed to an existing state-wide station network, we use the cross-validation scheme leave-one-station-out (LOO), where each station is left out exactly once for the test set. This is the most representative setting for the model deployment on AgWeatherNet.

Mean RMSE for this setting is averaged over all stations' test RMSE values and reported in Table 3, along with the 95% CI, for several time horizons and the aggregate of all 48 hours. EGP-GNN achieves the best performance at each horizon over all stations' RMSE values. EGP-GNN also has a smaller relative spread on the confidence intervals. We can see that the anchor seeding still improves performance over all baselines in the LOO setting, over all time horizons. This suggests that the anchor remains important for prediction accuracy, even with imperfect seeding of $y_i(T-1)$. A count of station best performances is shown in the Appendix, and a comparison with GraphWaveNet where GraphWaveNet uses $y_i(T-1)$ as a future covariate. Our model performs substantially better at the short horizons. Accurate short term forecasts (1 h) are just as important as longer horizons for the AWN decision support tools. These forecasts can make the difference between preparedness for a suitable time window, and delaying appropriate measures for crops.

4.2 Cluster Cross-Validation

Now, we investigate the more challenging problem of measuring the capacity for generalization to unseen regions. We set up a cluster-based partitioning of the data to define the cross-validation folds. The data is clustered into 6 groups using k-means clustering (Lloyd 1982) on latitude/longitude locations. We use four clusters for training, one cluster for validation, and one cluster for testing. As shown in Figure 2, there is no overlap of stations or time across data splits.

Table 3: Leave-one-station-out (LOOCV) test RMSE, mean ± 95% CI averaged across the 42 stations. Each station is held-out exactly once. All Horizons averages over all time horizons from 1 h through 48 h. Lower is better.

Forecast Horizon	LSTM	DKL-LSTM	GraphWaveNet	EGP-GNN	EGP-GNN (RF seed)
1 h	1.412 ± 0.130	1.407 ± 0.125	1.410 ± 0.121	$\mathbf{0.886 \pm 0.079}$	1.240 ± 0.108
24 h	1.533 ± 0.128	1.500 ± 0.128	1.509 ± 0.127	$\mathbf{1.417 \pm 0.116}$	1.456 ± 0.124
48 h	1.570 ± 0.129	1.518 ± 0.129	1.529 ± 0.130	$\mathbf{1.481 \pm 0.119}$	1.499 ± 0.123
All Horizons	1.527 ± 0.129	1.495 ± 0.129	1.506 ± 0.126	$\mathbf{1.418 \pm 0.117}$	1.450 ± 0.124

Table 4: Cluster-based partitioning (cluster CV) test RMSE, mean ± 95% CI across the 6 spatial folds. Results at 1, 24, and 48 h are pooled over each fold's held-out stations. All Horizons averages each fold's held-out stations equally and averages over all time horizons from 1 h through 48 h. Lower is better.

Forecast Horizon	LSTM	DKL-LSTM	GraphWaveNet	EGP-GNN	EGP-GNN (RF seed)
1 h	1.545 ± 0.361	1.697 ± 0.574	1.467 ± 0.174	$\mathbf{0.904 \pm 0.203}$	1.291 ± 0.169
24 h	1.692 ± 0.457	1.755 ± 0.567	1.557 ± 0.235	$\mathbf{1.479 \pm 0.205}$	1.535 ± 0.207
48 h	1.729 ± 0.481	1.766 ± 0.546	1.581 ± 0.241	$\mathbf{1.552 \pm 0.203}$	1.587 ± 0.212
All Horizons	1.611 ± 0.389	1.638 ± 0.407	1.518 ± 0.209	$\mathbf{1.445 \pm 0.196}$	1.485 ± 0.172

Table 5: Ablation study on the cluster-based CV, test RMSE as mean ± 95% CI across the 6 spatial folds. Results at 1, 24, and 48 h are pooled over each fold's held-out stations. All Horizons averages each fold's held-out stations equally and averages over all time horizons from 1 h through 48 h. Lower is better.

Forecast Horizon	−cnn	−gate	−anchor	EGP-GNN
1 h	0.937 ± 0.182	0.916 ± 0.211	1.422 ± 0.174	$\mathbf{0.904 \pm 0.203}$
24 h	1.507 ± 0.197	1.492 ± 0.252	1.621 ± 0.251	$\mathbf{1.479 \pm 0.205}$
48 h	$\mathbf{1.552 \pm 0.202}$	1.554 ± 0.237	1.679 ± 0.280	$\mathbf{1.552 \pm 0.203}$
All Horizons (1–48h)	1.510 ± 0.172	1.481 ± 0.214	1.572 ± 0.243	$\mathbf{1.445 \pm 0.196}$

Table 4 presents mean test RMSE results over all folds and various forecast horizons, along with the 95% confidence intervals (CI) for the mean RMSE. We can see that EGP-GNN again improves performance at each time horizon over the baselines, and generally has a tighter bound on the CI. On the cluster CV, RF seeding performance exceeds baselines on all time horizons except 48 h, which remains competitive with GraphWaveNet, with All Horizons still an improvement. A per-fold breakdown is available in the Appendix.

4.3 Ablations

We perform an ablation study on several components of EGP-GNN to investigate the RMSE impact of each. We ablate one-at-a-time: the CNN features $\mathbf{D}_i$, the learnable edge gate $\mathbf{g}_{ij}(t)$, and the learnable anchor term in the decoder $\lambda_i(\tau)$. The ablations use the same cluster-based cross-validation as Section 4.2 and are summarized in Table 5.

The full EGP-GNN model performs best at each horizon in terms of RMSE, with $\lambda_i(\tau)$ having the largest impact on forecasts. Removing the edge-gate widens the 95% CI, hurting consistency, while slightly harming the mean. Removing autoencoder terrain profiling via the CNN affects the 1 h and 24 h forecasts. We investigated alternative formulations for the temporal encoder and decoder in our model. These include temporal convolutional networks (Lea et al. 2017) for the encoder, cross-attention (Vaswani et al. 2017) and non-autoregressive decoders without the anchoring formulation, and adding more context into the decoder. In general, these changes harm the performance marginally. The temporal model for the encoder is relatively interchangeable, but affects the consistency of the model across folds. For uniformity, we use the GRU models for both encoder and decoder.

5 Deployment and Conclusion

EGP-GNN will be further validated and ported to a production environment for users on the AgWeatherNet portal (AgWeatherNet 2026). This portal provides frost risk forecasts, advisory on if mixing air helps or aggravates conditions on a given frost day, and pesticide spray advisories, both of which make use of temperature inversion forecasts directly. Initially, this model will operate in a testing capacity where it will generate inversion forecasts for existing tower stations, and test specifically on tower stations that were installed after August of 2025. This setting will yield valuable guidance and steps towards forecasting on non-tower stations, which the current AWN system does not provide, and for which benchmarking is difficult. We also plan to scale up our model to forecast at higher resolutions (5 to 15 min intervals).

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